SONVI ASSIS NORONHA

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Professional Summary

Machine Learning Engineer specializing in Computer Vision and Deep Learning with experience in hybrid CNN–Transformer architectures, feature fusion systems, and biometric recognition models. IEEE published researcher skilled in embedding optimization, ArcFace loss, cross-validation, and deployment-ready AI systems using PyTorch and TensorFlow. Passionate about building scalable, high-performance intelligent systems.

Education

HKBK College of Engineering, Bengaluru 2022 – 2026
Bachelor of Engineering in Artificial Intelligence and Machine Learning

Projects

SI-FF-ArcNet: Identical Twin Face Recognition (IEEE Published) 2025
Python, PyTorch, Swin Transformer, InceptionV4, ArcFace Loss, OpenCV

- Designed hybrid deep learning architecture integrating Swin Transformer and InceptionV4 using multi-scale feature fusion.
- Built 512-D ArcFace embedding pipeline with 5-fold cross-validation and advanced augmentation.
- Achieved 99.96% validation accuracy and 98.20% F1-score across 50 epochs.
- Implemented cosine similarity–based open-set recognition for biometric authentication.

Movie Recommendation System Dec 2024
Python, Pandas, Scikit-learn

- Built content-based recommendation engine using TF-IDF vectorization and cosine similarity.
- Achieved 85% recommendation accuracy through similarity scoring.

Real-Time Semaphore Detection System Aug 2025
Python, OpenCV, MediaPipe, Flask, Jetson Nano

- Developed real-time pose-based signal recognition system.
- Deployed on NVIDIA Jetson Nano for efficient edge inference.

Technical Skills

Programming	Python, SQL, Julia, R (Basics), Java (Basics), C (Basics)
Deep Learning	PyTorch, TensorFlow, CNN, Transformers, ArcFace, Model Optimization
Computer Vision	OpenCV, MediaPipe, Image Processing
Data Tools	Pandas, NumPy, Power BI
Tools	Git, GitHub, Jupyter, Google Colab, VS Code, Jetson Nano
Concepts	Cross-Validation, Feature Fusion, Open-Set Recognition, Hyperparameter Tuning

Certifications

- NPTEL (Elite) – Artificial Intelligence in Industrial Engineering (2025)
- NPTEL – Cloud Computing and Distributed Systems (2025)
- Generative AI & LLM Course – RAG, Embeddings, Prompt Engineering (2024)
- Infosys Springboard – MongoDB Developer (2024)
- Great Learning – Data Visualization with Power BI (2024)